

Concurrent and Distributed Systems

Lab 8 – Client/server applications with Java socket programming

Materials

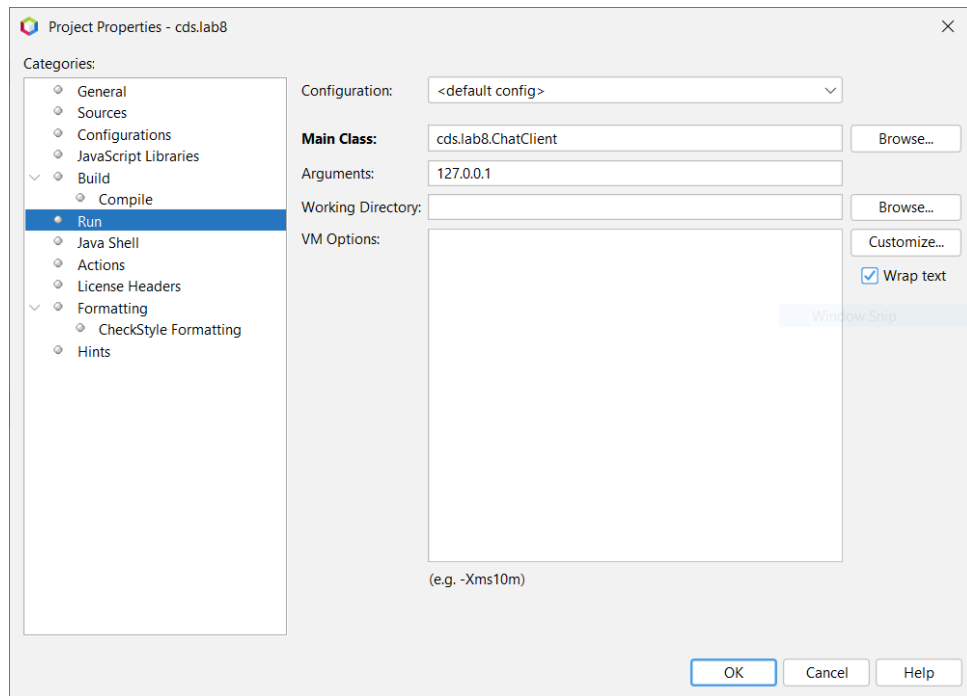
Project with examples (from <https://cs.lmu.edu/~ray/notes/javanetexamples/>): **cds.lab8.zip**

- A trivial date server (**DateServer**) and client (**DateClient**), illustrating **simple one-way communication**. The server sends data to the client only.
- A capitalize server (**CapitalizeServer**) and client (**CapitalizeClient**), illustrating **two-way communication**, and **server-side threads** to more efficiently handle multiple connections simultaneously.
- A two-player tic tac toe game (**TicTacToeServer** and **TicTacToeClient**), illustrating a server that needs to **keep track of the state of a game**, and inform each client of it, so they can each update their own displays.
- A multi-user chat application (**ChatServer** and **ChatClient**), in which a server must **broadcast** messages to all of its clients.

Lab session

Activity 1 – Run the client/server example applications

Import the project **cds.lab8.zip** in NetBeans. Check out the code of the client/server example applications and run them. For each of them, you need to first start the server and then the client(s). You can run them on the same machine – in a real distributed system they would run on different machines. To start the client you need to specify the IP address of the server (127.0.0.1, i.e. the local host) as a command line argument – in NetBeans you can do this in the project properties (right click on the project name -> Properties):



Recall that ports from 49152 to 65535 can be used for anything you want – these are the ports used in the examples. You can find more details on the example applications at:

<https://cs.lmu.edu/~ray/notes/javanetexamples/>

Activity 2 – Improve Chat application to distinguish system messages from chat messages

Enhance the protocol so that it distinguishes system messages (such as people joining and leaving), from chat messages, then enhance the client so that it uses color and fonts to highlight the different kinds of messages.

Activity 3 – Improve Chat application to send personal messages

Extend the protocol so that a message can be sent only to a specific user, then enhance the client to incorporate this new functionality.

Resources

- Ray Toal, Java Socket Programming Examples, <https://cs.lmu.edu/~ray/notes/javanetexamples/>